Quiz — 1.4.3 Boolean Algebra

Name: ____ Date: _ ***Mark:*** ___ / 30

OCR H446 · Component 01 · 1.4.3 — show all working

Section A — Multiple choice (8 marks)

1. For which input combination does a 2-input **AND** gate output 1? A. A=0, B=0 B. A=1, B=0 C. A=0, B=1 D. A=1, B=1 Your answer: ☐

2. Which logic gate outputs 1 **only when its two inputs differ**? A. AND B. OR C. XOR D. NOR Your answer: ☐

3. Which two gates are described as **universal** (any circuit can be built from them alone)? A. AND and OR B. NAND and NOR C. XOR and NOT D. OR and NOT Your answer: ☐

4. De Morgan's law states that `¬(A + B)` is equivalent to: A. `¬A + ¬B` B. `¬A · ¬B` C. `A · B` D. `¬(A · B)` Your answer: ☐

5. Which Boolean identity is `¬(¬A) = A`? A. Idempotent B. Absorption C. Double negation D. Complement Your answer: ☐

6. In a **half adder**, the Sum output is given by: A. `A · B` B. `A + B` C. `A ⊕ B` D. `¬(A · B)` Your answer: ☐

7. What is the role of a **D-type flip-flop**? A. To add two bits together B. To store one bit of data on a clock edge C. To invert its input D. To convert binary to hexadecimal Your answer: ☐

8. When grouping 1s on a Karnaugh map, groups must be: A. Always exactly size 3 B. As small as possible C. Powers of two (1, 2, 4, 8…) and as large as possible D. Only ever a single cell Your answer: ☐

Section B — Calculations & short answer (17 marks)

9. Complete the truth table for the expression **Q = ¬(A · B) + (A · C)**. **[5]**

A	B	C	A·B	¬(A·B)	A·C	Q
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

10. Use **De Morgan's law** (and double negation) to simplify `¬(¬A + B)` to an expression with no outer bar. Show each step. **[3]**

Working space:

11. A **half adder** receives inputs **A = 1** and **B = 1**. State the **Sum** and **Carry** outputs and the Boolean expression for each. **[3]**

Working space:

12. Complete this truth table for a **half adder** for all input combinations. **[3]**

A	B	Sum ($A \oplus B$)	Carry ($A \cdot B$)
0	0		
0	1		
1	0		
1	1		

13. Simplify the expression `A · (A + B)` using a Boolean identity, and name the law used. **[3]**

Working space:

Section C — Exam-style question (5 marks)

14. A **full adder** adds three input bits: A, B and a carry-in (Cin), producing a Sum and a carry-out (Cout).

(a) Give the Boolean expression for the **Sum** output of a full adder. **[1]**

(b) Give the Boolean expression for the **Cout** output of a full adder. **[2]**

(c) Explain how a full adder differs from a half adder, and why full adders are needed to add multi-bit binary numbers. **[2]**

Working space: